

NETHEESWARAN A

DATA SCIENTIST | AI & MACHINE LEARNING

netheeswarana@gmail.com | [+353 894001009](tel:+353894001009) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Data Scientist with 3+ years of experience working with data analysis, machine learning, and large datasets to solve business problems. Skilled in Python and SQL, with experience building data pipelines, performing analysis, and delivering insights through dashboards. Comfortable working with cross-functional teams and turning data into practical, usable outcomes.

Technical Skills

- **GenAI Engineering:** Multi-Agent Systems, RAG Architectures, ReAct/CoT/Self-Ask Patterns, Prompt Engineering, Guardrails & Safety.
 - **LLM Providers:** OpenAI, Anthropic, Google Gemini, Hugging Face (Llama/Mistral)
 - **Backend Engineering:** Python, FastAPI, REST APIs, Microservices
 - **Machine Learning:** NLP, Embeddings, Fine-Tuning, Model Evaluation, Hugging Face Transformers
 - **DevOps & Infra:** Docker, Kubernetes, GitHub Actions (CI/CD), Monitoring/Logging (MLflow, LangSmith)
 - **Workflows:** Airflow, Prefect
 - **Data Engineering:** SQL, PostgreSQL, Snowflake, Data Lakes, ETL/ELT Pipelines
 - **Cloud:** AWS (EC2, S3), Azure Synapse/Data Lake, GCP (basic)
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Professional Experience

TATA Consultancy Services (TCS) | Data Scientist | Bangalore | Mar 2022 – Aug 2024

Responsibilities:

- Built machine learning models to analyse large-scale datasets (10M+ rows), applying statistical modelling and data analysis techniques to identify trends and support product and business decision-making.
- Developed statistical models and performed exploratory data analysis (EDA), using Python to uncover performance drivers and improve prioritisation accuracy by 18%.
- Designed data pipelines and preprocessing workflows, improving data quality and enabling reliable data integration for downstream analytics and machine learning components.
- Analysed structured and unstructured data using SQL and Python, generating actionable insights to enhance product performance and support data-driven development.
- Built scalable data pipelines for training, evaluation, and batch inference, contributing to production-ready analytics components and efficient data processing.
- Developed dashboards and visualisations using Power BI and Python, delivering clear reporting and insights for stakeholders and supporting decision-making processes.
- Collaborated within cross-functional teams, translating business requirements into analytical solutions and contributing to agile development workflows.

Projects:

Predictive Financial Risk & Forecasting Model

- Built a supervised ML model to analyze and prioritise 500K+ records, improving accuracy by 18% and supporting data-driven decision-making.
- Engineered features from large-scale Snowflake datasets (10M+ rows), identifying patterns and trends to improve model performance.
- Designed data validation workflows, reducing data errors by 30% and ensuring reliable inputs for analysis and modeling.
- Deployed the model on Azure with MLflow tracking and CI/CD pipelines, enabling scalable batch processing (100K+ records/run).

PaaS Insurance RAG Application

- Developed an AI-driven system to analyze unstructured documents, enabling users to extract insights from policy, claims, and regulatory data.
- Built data pipelines to process and structure unstructured text, improving accessibility and usability of information.
- Optimized retrieval and ranking logic, reducing hallucinations by 40% and improving output reliability.
- Evaluated system performance using retrieval and language model metrics to ensure accuracy and consistency.

LLM Fine-Tuning for Investment Question Answering

- Developed a domain-specific model to improve question answering accuracy on investment and policy datasets.
- Created structured datasets from historical data to enhance model performance and domain understanding.
- Evaluated model performance using standard NLP metrics and benchmark comparisons.
- Integrated the model into a retrieval-based system, improving response accuracy by 30% for complex queries.

Cloud Data Analyst - Snowflake | Bangalore | Nov 2020 – Mar 2022

- Performed large-scale data validation for Azure Teradata to Snowflake migration (12TB, 420+ tables), ensuring high-quality, consistent datasets for analytics and machine learning workflows.
- Built automated validation and testing frameworks using SQL and Python (pytest, ADF, GitHub Actions), improving data quality, reliability, and CI/CD-driven data processes.
- Designed Power BI dashboards to monitor data quality KPIs, providing actionable insights and improving visibility into system performance and reporting accuracy.
- Authored technical documentation and governance playbooks, supporting data quality standards, reproducibility, and effective collaboration across teams.

Personal Projects

AI Medical Assistant - Tech stack - Python, Pinecone, OpenAI, Langchain, Groq, HuggingFace, Flask - [GitHub](#)

- Developed an AI Medical chatbot that answers medical queries by loading medical PDF into the Pinecone vector database and utilising Groq for NLP.
- Deployed on AWS EC2 with GitHub Actions for CI/CD.

AI Career Helper - Tech stack - Python, FAISS, OpenAI, LangChain, Streamlit - [GitHub](#)

- Designed and developed an AI-powered assistant to guide users on careers in AI, Machine Learning, and Data Science.
- The system extracts and embeds content from industry career guides and interview prep documents into a FAISS vector index, enabling semantic search and context-aware query resolution.

Education

Masters in Data and Computational Science | Sep 2024 - Aug 2025 | Dublin, Ireland

University College Dublin - Grade 2:1 - 3.6/4.2

Bachelors in Electronics and Communication Engineering | July 2016 - Mar 2020 | Erode, India

Kongu Engineering College - CGPA: 8.0/10

Certifications & Courses

MLOps (Coursera) | Data Science Specialization (Coursera) | Python for ML & Data Science (Udemy) |

Snowflake Complete Masterclass (Udemy) | SQL Advanced (Hackerrank) | Microsoft Power BI (Udemy)